

GIK ScreenPanel placement — 120 cm configuration

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GIK ScreenPanel placement at the left-wall reflection point — 120 cm configuration

Scope and answer

This note answers two connected questions for the 120.blue room:

1. Where exactly is the left loudspeaker's first reflection point on the concrete left wall?
2. With the hinged GIK ScreenPanel standing as an open book, should its longitudinal centre/hinge be placed on that point, or should the screen be shifted farther from the loudspeaker while still covering the point?

It also records the measured result of the new treatment order on the open right side: GIK panels S3/S4 first, prolonging the 1.75 m half-wall, followed by foam panels F1/F2.

The exact left-wall point for the left loudspeaker is:

197.22 cm from the front wall and 117.01 cm above the floor.

For a ScreenPanel opened to approximately 90 degrees, do **not** move the hinge far enough that this wall point merely falls at the projected edge of the screen. The ray then passes through the open V without touching either absorber leaf. The recommended 90-degree placement is:

Put the hinge/longitudinal centre at 199 cm from the front wall: about 2 cm farther from the speaker than the exact wall point.

This small shift makes the outgoing ray cross approximately the middle of the rear leaf instead of meeting the hinge. A practical safe range is **198.2 to 200.2 cm**, assuming a 10 cm margin from both the hinge and outer leaf edge.

The intuition behind moving the panel farther from the speaker is therefore partly right: the ray should meet absorptive face, not the hinge. But a large translation is safe only when the ScreenPanel is opened much flatter. If more physical clearance is required, change the opening angle as well as the position.

1. Geometry and exact reflection point

The coordinates come from [roomgeom.py](#), based on the tape survey of 2026-08-10. Coordinates are:

- x : distance from the left wall;
- y : distance from the front wall;
- z : height above the floor.

The relevant points in metres are:

item	coordinate (x, y, z)
left tweeter S	(0.680, 1.200, 1.200)
listening point M	(2.050, 4.300, 1.080)
image of the left tweeter S'	(-0.680, 1.200, 1.200)

The specular path is obtained by reflecting the source through the wall and drawing the straight line from S' to M . Its intersection with $x = 0$ is:

$$P = (0, 1.972161, 1.170110) \text{ m.}$$

The direct path is 3.391357 m and the reflected path is 4.132469 m. Their 0.741112 m difference corresponds to **2.16068 ms** at 343 m/s. This is the exact version of the earlier rounded **1.97 m / 2.2 ms** result in the [room panel map](#).

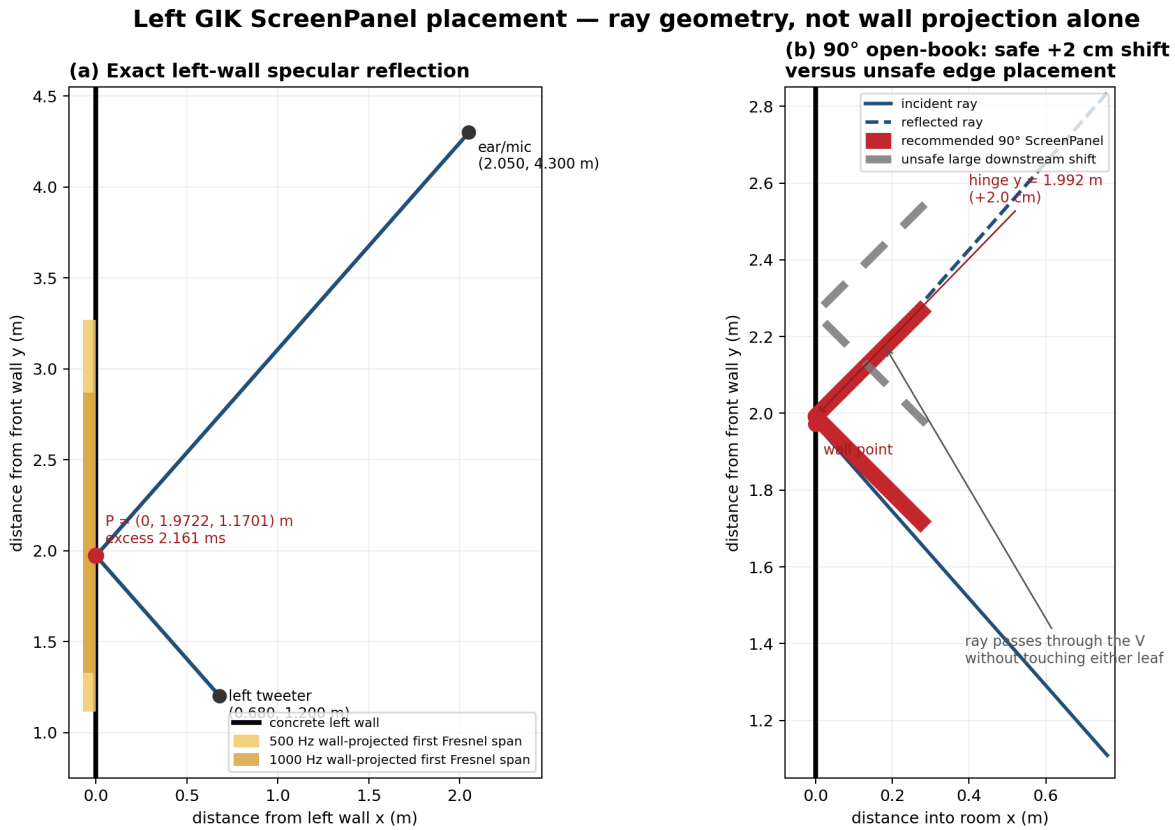


Figure 1: Exact reflection ray and the open-book interception geometry.

The height is already appropriate for a floor-standing 183 cm ScreenPanel: the point is 117 cm high, leaving approximately 117 cm below it and 66 cm above it.

2. Why projected wall coverage is not sufficient

The legacy GIK ScreenPanel is 815 mm wide when unfolded and has two hinged halves, each approximately **407.5 mm** wide. It is 1830 mm high and about 76 mm thick. Those dimensions and its intended use at reflection points are given by GIK's [ScreenPanel product description](#).

In plan view, assume:

- the hinge touches the concrete wall;
- the two equal leaves open symmetrically into the room;
- a is the included opening angle between the leaves;
- $w = 0.4075$ m is one leaf's width;
- the centre/hinge is shifted by Δy toward the listener, away from the speaker.

Each endpoint is then:

$$d = w \sin \left(\frac{180^\circ - a}{2} \right)$$

into the room, and

$$h = w \cos \left(\frac{180^\circ - a}{2} \right)$$

along the wall from the hinge.

The incident and outgoing rays both have a plan-view slope of approximately 1.1355 m along the wall per metre into the room. For a downstream hinge shift, the maximum shift at which either ray only just touches a leaf endpoint is:

$$\Delta y_{\text{limit}} = |h - 1.1355d|.$$

Putting the ray at an endpoint is fragile. The recommendation below instead places the intersection at the middle of the active leaf:

$$\Delta y_{\text{recommended}} = \frac{1}{2} \Delta y_{\text{limit}}.$$

The shaded region in the following plot retains at least 10 cm of absorber between the ray and either leaf edge. The scaled plan-view sketches below the graph show the included 90, 120, 150 and 180 degree openings explicitly; the black line is the wall, the square is the hinge and the red segments are the two absorbing leaves.

There is a critical opening near **97.3 degrees** where a leaf is nearly parallel to the ray. Around that angle almost any translation makes the ray miss the panel. That is why the longitudinal wall projection alone gives the wrong answer for an open-book absorber.

3. Recommended positions by opening angle

included opening	depth into room	projected half-span	safe downstream shift	recommended shift	recommended hinge y	nearest endpoint y
90 deg	28.8 cm	28.8 cm	1.0-3.0 cm	2.0 cm	1.992 m	1.704 m
120 deg	20.4 cm	35.3 cm	3.0-9.2 cm	6.1 cm	2.033 m	1.680 m
150 deg	10.5 cm	39.4 cm	6.7-20.7 cm	13.7 cm	2.109 m	1.716 m
180 deg, flat	0 cm	40.8 cm	10.0-30.8 cm	20.4 cm	2.176 m	1.768 m

The nearest endpoint in the recommended 90-degree position is 50.4 cm behind the tweeter along the room-length direction and approximately **64 cm from the tweeter in plan**. Opening the panel flatter increases that plan clearance to about 68 cm at 120 degrees, 77 cm at 150 degrees and 89 cm when flat.

Practical decision

- **If the panel must remain self-supporting at about 90 degrees:** mark the hinge at **199 cm** from the front wall. Do not translate it substantially farther.
- **If more speaker clearance is important:** open it to about 150 degrees and put the hinge at **211 cm**, provided it remains mechanically secure.
- **If it can be held almost flat against the wall:** put the hinge at about **218 cm**. The exact reflection point then lies near the middle of the speaker-side half, not precariously at the outer edge.
- **Do not use the “point just covered by the edge” rule.** Even flat, it gives no tolerance for tape error, speaker toe-in, listener movement or the finite reflection footprint.

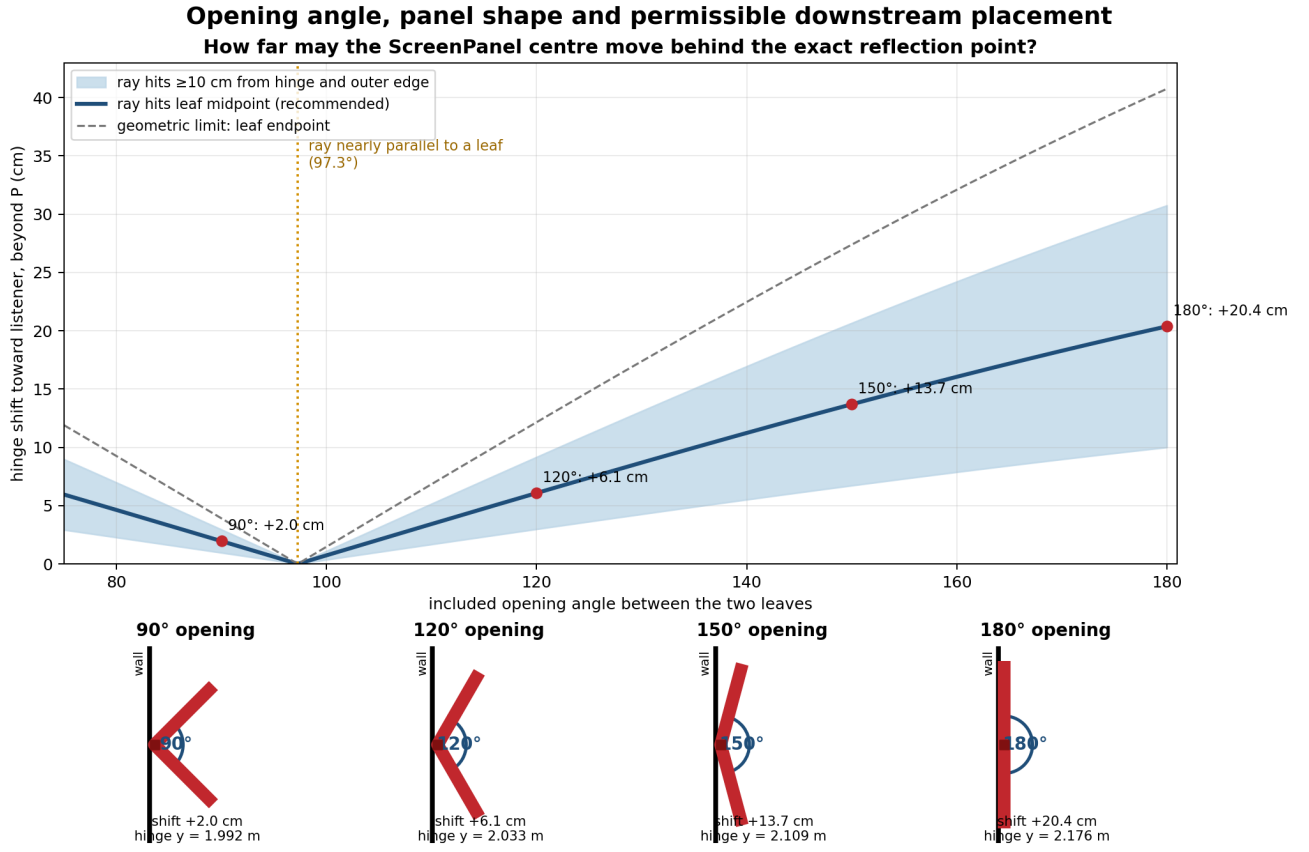


Figure 2: Permissible downstream shift as a function of ScreenPanel opening angle.

4. Finite reflection footprint

A single geometrical ray locates the centre of a reflection, but real audio occupies a finite first Fresnel zone. Solving the exact path-length condition

$$|S - Q| + |Q - M| = |S - P| + |P - M| + \frac{\lambda}{2}$$

for points Q on the concrete wall gives these wall-projected spans:

frequency	longitudinal span y	width along wall	vertical span z
250 Hz	0.821-3.826 m	3.005 m	0.043-2.281 m
500 Hz	1.114-3.271 m	2.157 m	0.404-1.927 m
1 kHz	1.328-2.869 m	1.540 m	0.641-1.695 m
2 kHz	1.493-2.588 m	1.095 m	0.800-1.537 m
4 kHz	1.619-2.396 m	0.776 m	0.910-1.429 m

The 81.5 cm ScreenPanel can cover the full longitudinal first Fresnel span only from roughly 4 kHz upward. It still attenuates lower-frequency reflection energy, but it cannot absorb the whole 500 Hz or 1 kHz footprint by itself. This reinforces two placement rules:

1. intercept the exact central ray on absorptive material, not on the hinge or edge;
2. do not sacrifice central coverage merely to gain a few centimetres of loudspeaker clearance.

Vertically, the floor-standing panel covers the full 1 kHz-and-up zone and almost all the 500 Hz zone; at 500 Hz only the uppermost approximately 10 cm lies above the 183 cm panel.

5. Measured result of the new right-side order

Two independent differential sets compare the traditional right-side order (foam first, then GIK) with the new order (GIK S3/S4 first, then foam):

- `../DRC-120.blue/foam.screens.opendoor.mdat`, measured 2026-08-17 with effectively the same microphone position;
- `../DRC-120.blue/120.blue.Rscreen.txts.boh/`, measured 2026-08-10 with a 3-4 cm microphone displacement between the new and trad measurements.

All response comparisons below use common 1/6-octave smoothing and remove only the median 500 Hz-5 kHz acquisition offset. REW stores frequency response and impulse response as separate measurement products, as described in its [measurement API documentation](#).

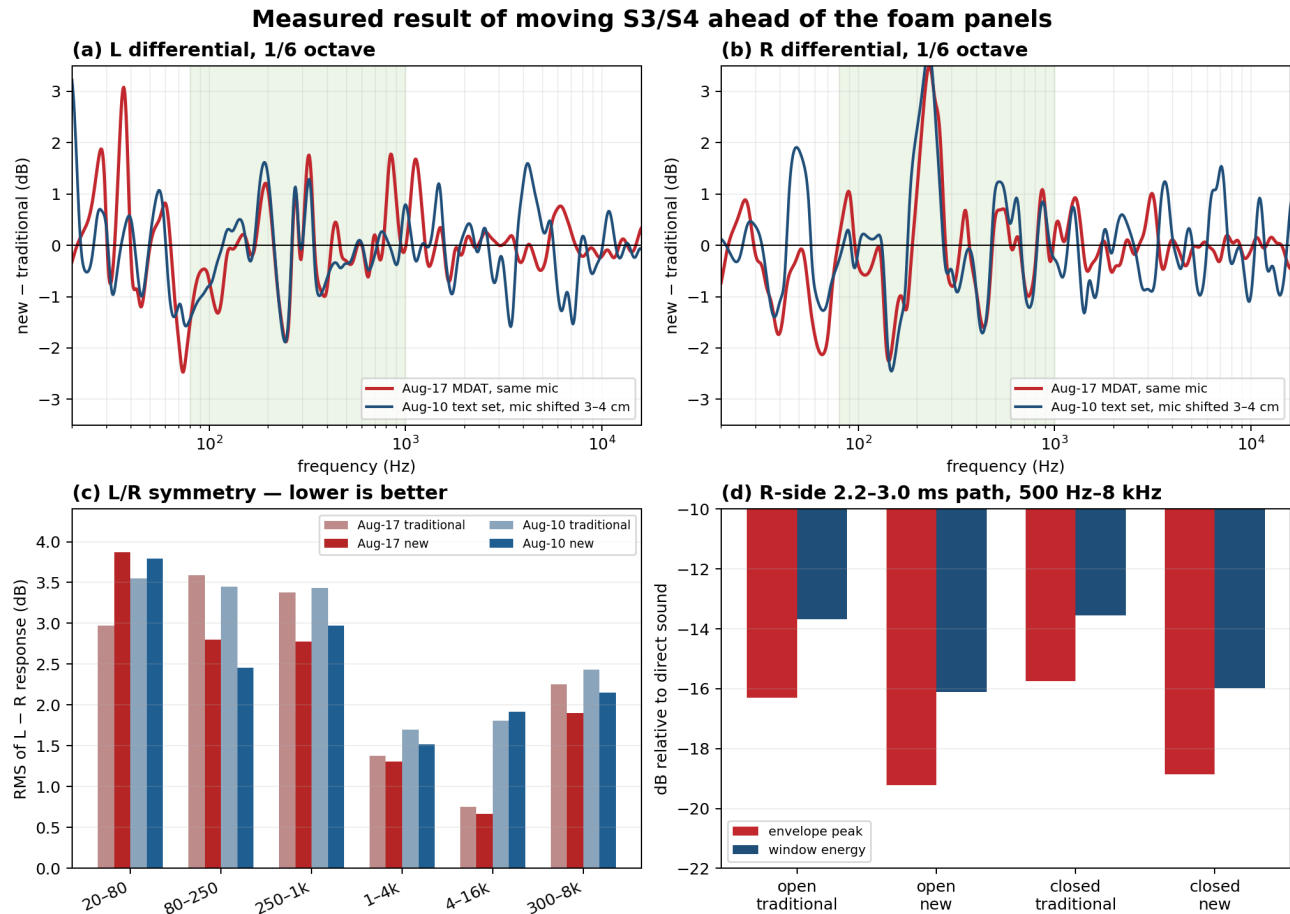


Figure 3: Frequency, symmetry and early-reflection results for the panel reorder.

5.1 Left/right response symmetry

The metric is the RMS of the smoothed $L - R$ response; lower is more symmetric.

band	Aug-17: traditional to new	Aug-10: traditional to new	conclusion
20-80 Hz	2.97 to 3.87 dB	3.55 to 3.79 dB	no bass benefit
80-250 Hz	3.59 to 2.80 dB	3.45 to 2.46 dB	strongest repeatable gain
250 Hz-1 kHz	3.38 to 2.78 dB	3.43 to 2.97 dB	moderate gain
1-4 kHz	1.38 to 1.31 dB	1.70 to 1.52 dB	small gain
4-16 kHz	0.75 to 0.67 dB	1.81 to 1.92 dB	not repeatable
300 Hz-8 kHz	2.25 to 1.90 dB	2.43 to 2.15 dB	modest overall gain

The two differential curves correlate at **0.90/0.92** for **L/R** over **80-250 Hz** and at **0.69/0.86** over **250 Hz-1 kHz**. Above 1 kHz the agreement largely disappears, consistent with the microphone displacement and comb-filter sensitivity. The credible audible change is therefore a modest improvement in lower-midrange L/R matching and phantom-centre stability, not a global tonal transformation.

This matches the earlier material argument: foam and GIK treatment may look geometrically symmetric, but they are spectrally very different in the 125-500 Hz band. See the the earlier treatment arithmetic.

5.2 The S3 reflection path

The right loudspeaker’s geometrical reflection point on the prolonged right wall lies at about 2.24 ms. In the measured 2.2-3.0 ms region, filtered from 500 Hz to 8 kHz:

door state	traditional peak	new peak	peak change	energy change
open	-16.30 dB	-19.23 dB	-2.93 dB	-2.43 dB
closed	-15.75 dB	-18.86 dB	-3.11 dB	-2.43 dB

This is the strongest path-specific result and validates putting S3 at the right-speaker mirror point. Total R-channel energy in the wider 2-5 ms window falls only about 0.6 dB because the much stronger floor reflection near 1.8 ms is unchanged.

5.3 S4, bass and the entrance door

- The expected L-to-S4 cross-room path near 8.5 ms does not fall repeatably: the two comparisons give approximately +0.6 and -0.2 dB peak changes.
- There is no treatment benefit below about 80 Hz. This is expected from two porous panels and the room’s modal behaviour.
- Opening the entrance door changes the response by less than about 0.2 dB above 63 Hz and by no more than roughly 0.4 dB over 40-63 Hz. A weak 35-50 Hz late-energy increase is marginal and, if real, is slightly more ringing rather than an improvement.

The door is therefore not an audible control variable at the 120 cm speaker position.

6. Qualification: the left-panel move is a confound

The 2026-08-17 measurement-group note says that the first left GIK panel was also moved closer to the left-speaker reflection point. Therefore the full new-versus-traditional response comparison is a comparison of the **combined layout**, not a perfectly isolated test of only the S3/S4 versus F1/F2 order.

That does not invalidate the path-specific S3 result: the approximately 3 dB reduction occurs on the R-side 2.2-3.0 ms path and repeats with the door open and closed. It does mean that broad L-channel changes should not be assigned solely to the right-side reorder.

7. Installation and verification procedure

1. Measure along the concrete left wall from the front-wall plane.
2. Mark the exact ray point at **197.2 cm**, height **117.0 cm**.
3. Choose and record the actual included angle between the ScreenPanel leaves.
4. Mark the hinge according to the table:
 - about **199 cm** at 90 degrees;
 - about **203 cm** at 120 degrees;
 - about **211 cm** at 150 degrees;
 - about **218 cm** if flat.
5. Confirm that the ray-facing fabric, not the wooden hinge/frame, occupies the intersection.
6. Do not move the microphone, loudspeaker, S2, S3/S4 or the door during the comparison.
7. Sweep the L loudspeaker at two candidate S1 angles/positions and compare:
 - the 1.5-3 ms ETC region;
 - 80 Hz-1 kHz L/R symmetry;
 - the response after common 1/6-octave smoothing.

The primary recommended test is **90 degrees at 199 cm versus 150 degrees at 211 cm**. That directly answers whether the extra physical clearance of the flatter setup costs any measurable reflection attenuation.

8. Rebuilding the figures and PDF

The figures are generated by:

```
python3 gik_screen_panel_placement-120cm.py
```

The measurement figure additionally requires `javaobj-py3` to decode the REW MDAT:

```
python3 -m pip install javaobj-py3
```

Build the PDF from this directory with:

```
pandoc GIK-SCREEN-PANEL-PLACEMENT-120cm.md \  
-o GIK-SCREEN-PANEL-PLACEMENT-120cm.pdf \  
--from=markdown+gfm_auto_identifiers \  
--pdf-engine=pdflatex \  
--include-in-header=pdf-header.tex \  
--toc --toc-depth=3 \  
--metadata title="GIK ScreenPanel placement and measured panel-order result - 120 cm configuration" \  
--metadata author="giacomo" \  
-V geometry:a4paper -V geometry:margin=2cm \  
-V fontsize=10pt -V colorlinks=true \  
-V linkcolor=RoyalBlue -V urlcolor=RoyalBlue -V toccolor=black
```